FINAL EXAM BRIDGING
SEMESTER I, SESSION 2020/2021

COURSE CODE	**:**	**IFC 1024**
COURSE NAME	**:**	**CHEMISTRY**
PROGRAMME	**:**	**BRIDGING SPACE**
DURATION	**:**	**3 HOURS**
DATE	**:**	**09 FEBRUARY 2021**

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DOSO.

2. ANSWER ALL QUESTIONS.

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This booklet consists of (16) printed pages including this page.

QUESTION 1 (15 MARKS)

a) Calculate the standard enthalpy of formation of acetylene, C_2H_2 from its elements :

$$2\ C\ (graphite) + H_2\ (g) \rightarrow C_2H_2\ (g)$$

The equations for each step and the corresponding enthalpy are :

$C\ (graphite) + O_2\ (g) \rightarrow CO_2\ (g)$	$\Delta H^o = -393$ kJ/mol ×2
$H_2\ (g) + \frac{1}{2} O_2\ (g) \rightarrow H_2O\ (l)$	$\Delta H^o = -285.8$ kJ/mol
$2\ C_2H_2\ (g) + 5\ O_2\ (g) \rightarrow 2H_2O\ (l) + 4\ CO_2\ (g)$	$\Delta H^o = -2598.8$ kJ/mol ÷2

(5 marks)

b) A quantity of 0.565 g of liquid hydrazine (N_2H_4) a pungent-smelling substance used as rocket fuel propellant, was burned in a constant-volume bomb calorimeter. The temperature rose from 19.28°C to 35.95°C. If the heat capacity of the bomb plus water was 10.17 kJ/°C, calculate the molar heat of combustion of naphthalene.

(4 marks)

c) A 37 g sample of an unknown metal at 79.0°C was placed in a constant-pressure calorimeter containing 50 g of water at 24.0°C. The final temperature of the system was found to be 26.4°C. Assuming that no heat is lost to the surroundings, calculate the specific heat of the metal. [$S(H_2O)$ = 4.184 J/g.°C]

(4 marks)

d) Determine whether each process is exothermic or endothermic :

i. Natural gas burning on a stove
ii. Water evaporating to surrounding

(1+1 marks)

QUESTION 2 (10 MARKS)

a) Given a reaction :2 $H_2(g)$ + 2 $NO(g) \rightarrow N_2(g) + 2H_2O(l)$

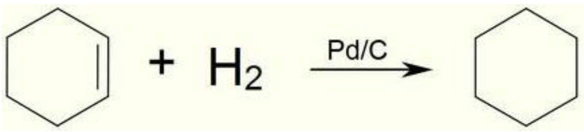
Experiment	**[H_2] (M)**	**[NO] (M)**	**Initial Rate (M/s)**
1	2.0×10^{-3}	5.0×10^{-3}	1.3×10^{-5}
2	2.0×10^{-3}	10.0×10^{-3}	5.0×10^{-5}
3	4.0×10^{-3}	10.0×10^{-3}	10.0×10^{-5}

Based on the reaction and table above :

i. Determine the rate law expression
ii. Calculate the rate constant, k

(7+1 marks)

b) Hydrogenation of cyclohexene form a cyclohexane in the presence of Pd/C as catalyzer. Cyclohexane is used in various solvent applications. It is volatile with a petroleum-like odor. The reaction of hydrogenation is given below at 75°C via the following first order reaction.



If the rate constant is 5.36×10^{-4} s^{-1}, calculate the half-life of the reaction in minute.

(2 marks)

QUESTION 3 (15 MARKS)

a) Consider the following equilibrium systems. Predict the direction of the net reaction in each case as a result of increasing the pressure (decreasing the volume) on the system at constant temperature. The reactants are in a cylinder fitted with a movable piston.

i. A (s) $\rightleftharpoons$ 2B (s)
ii. A (s) $\rightleftharpoons$ B (g)
iii. 2A (g) $\rightleftharpoons$ B (g)

(1+1+1 marks)

b) A natural source of nitrogen oxides (NO) occurs from a lightning stroke. The very high temperature in the vicinity of a lightning bolt causes the gases oxygen and nitrogen in the air to react to form nitrogen oxide. The nitrogen oxide then reacts very quickly with more oxygen to form nitrogen dioxide as shown below :

$$2\ NO\ (g) + O_2(g) \rightleftharpoons 2\ NO_2\ (g)$$

At equilibrium with 90°C, there are [NO] = 13.6 x 10^{-2} M, [O_2] = 0.54 M, and [NO_2] = 0.41 M. Calculate the equilibrium constant, Kc and then Kp

(4 marks)

c) Carbon dioxide (CO_2) a greenhouse gas will dissolved in seawater according to the reaction given below. Formation of calcium carbonate, $CaCO_3$ may ultimately reduce earth temperature by reducing the concentration of CO_2 in the air. Explain this phenomena in terms of Le Chatelier principle.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+\ HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$$

(2 marks)

d) At 445K, the equilibrium constant for the breakdown of NO_2 is 1.8x10^{-6}. Suppose 0.5 mol of NO_2 is placed in a 1.00 dm^3 reaction vessel. Calculate the concentration of each species at equilibrium.

The equation for the reaction is : $2\ NO_2(g) \rightleftharpoons 2\ NO(g) + O_2(g)$

(6 marks)

QUESTION 4 (15 MARKS)

a) Calculate the concentration of 100 mL potassium hydroxide (KOH) that has a pH of 13.

(2 marks)

b) Consider the mixing of potassium fluoride (KF) into 1.50 L of 0.050 M hydrofluoric acid (HF) to form a buffer solution (Ka of HF = 6.6×10^{-4})?

i. Assuming that no volume change occurs when the KF is added, Calculate the number of moles of KF need to be added into the solution to form a buffer solution with pH of 5.5

ii. Calculate the final pH if 10 mL of 0.5 M HCl is added into the solution

(4+5 marks)

c) A 0.100 M solution of chloroacetic acid ($ClCH_2COOH$) has a K_a value of 1.3×10^{-3}. Calculate the percentage ionization of the acid.

(4 marks)

4) (a) $pOH = 1 = -\log[OH^-]$ $\qquad [KOH] = [OH^-] = 0.1M$

$-\log[OH^-] = 1$

$[OH^-] = 10^{-1}$

$= 0.1M$

(b) (i) weak acid : HF

conjugate base : F^-

salt of conjugate base : KF

mol of HF = 1.5×0.05

$= 0.075 mol$

$5.5 = -\log(6.6\times10^{-4}) + \log\left(\frac{[c.base]}{0.075}\right)$

$[c.base] = 15.65 mol$

(ii) mol of HCl added = $0.01 \times 0.5 = 5\times10^{-3} mol \equiv [H^+]$

$H^+(aq)$	+ $F^-(aq)$	$\rightarrow HF(aq)$
5×10^{-3}	15.65	0.075
-5×10^{-3}	-5×10^{-3}	$+5\times10^{-3}$
0	15.645	0.08

$pH = -\log(6.6\times10^{-4}) + \log\left(\frac{15.645}{0.08}\right)$

$= 5.47$

(c) $ClCH_2COOH \rightleftharpoons ClCH_2COO^- + H^+$

$ClCH_2COOH$	$ClCH_2COO^-$	H^+
0.1	0	0
$-x$	$+x$	$+x$
$0.1-x$	x	x

$1.3\times10^{-3} = \frac{x^2}{0.1}$

$x = 0.0114$

% ionization $= \frac{0.0114}{0.1} \times 100$

$= 11.4\%$

QUESTION 5 (15 MARKS)

a) Complete and balance the following equation in acidic solution :

$$MnO_4^- (aq) + Br^- (aq) \rightarrow MnO_2 (s) + BrO_3^- (aq)$$

(5 marks)

b) Give TWO (2) functions of salt bridge in a voltaic cell setup.

(2 marks)

c) Consider the redox reaction below :

$$Fe^{3+} (aq) + Al (s) \rightarrow Fe (s) + Al^{3+} (aq)$$

i. Using standard reduction potential, calculate the standard EMF for the reaction.
ii. Write the half reactions equations
iii. Write the short-hand notation for the voltaic cell above

(1+2+1 marks)

d) Consider a concentration cell (voltaic cell) at temperature of 25ºC. Calculate the voltage produced by the cell. The setup is as below :

$$|Cr(s)|Cr^{3+}(0.5\ M)||Pb^{2+}(2.1\ M)|Pb(s)|$$

(4 marks)

5) (a) oxi : $Br^-(aq) \rightarrow BrO_3^-(aq)$

red : $MnO_4^-(aq) \rightarrow MnO_2(s)$

oxi : $Br^-(aq) + 3H_2O(l) \rightarrow BrO_3^-(aq) + 6H^+$

$Br^-(aq) + 3H_2O(l) \rightarrow BrO_3^-(aq) + 6H^+ + 5e^-$

red : $MnO_4^-(aq) + 4H^+ \rightarrow MnO_2(s) + 2H_2O$

$MnO_4^-(aq) + 4H^+ + 3e^- \rightarrow MnO_2(s) + 2H_2O$

QUESTION 6 (10 MARKS)

a) Figure 1 shows a structural formula of remdesivir, an antiviral drugs which in the past was touted as a possible remedy for Covid-19. Draw and list FOUR (4) functional groups from the structure.

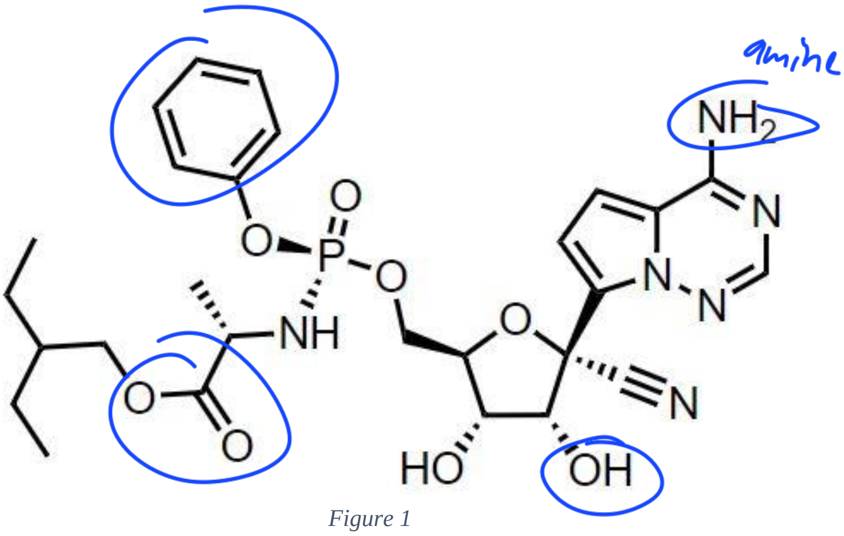
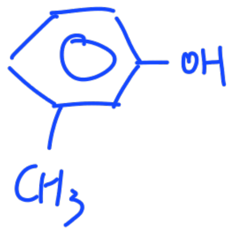


Figure 1

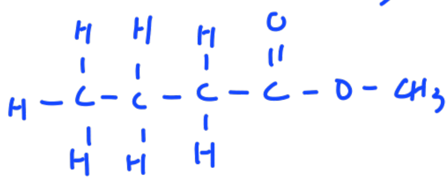
(4 marks)

b) Draw the structure of the following molecules :

i. m-methylphenol



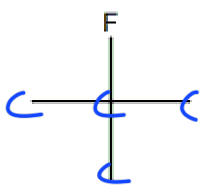
ii. ethyl butanoate



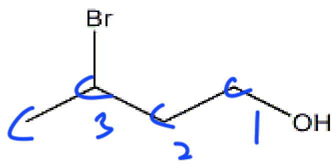
(1+1 marks)

c) Name the following structures :

i.



ii.



3-bromobutanol

(1+1 marks)

d) Draw TWO (2) possible structures of C_4H_6.

(2 marks)

The Periodic Table

1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VIIIB	10	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1 **H** 1.01																	2 **He** 4.00
3 **Li** 6.94	4 **Be** 9.01											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
11 **Na** 22.99	12 **Mg** 24.31											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.07	17 **Cl** 35.45	18 **Ar** 39.95
19 **K** 39.1	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.88	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.85	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.39	31 **Ga** 69.72	32 **Ge** 72.61	33 **As** 74.92	34 **Se** 78.96	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.94	43 **Tc** (98)	44 **Ru** 101.07	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.29
55 **Cs** 132.9	56 **Ba** 137.3	57 **La*** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.9	75 **Re** 186.2	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 **Ac^** (227)	104 **Rf** (261)	105 **Db** (262)	106 **Sg** (263)	107 **Bh** (264)	108 **Hs** (265)	109 **Mt** (268)	110 **Ds** (271)	111 **Rg** (272)							

*	58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3	65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
^	90 **Th** 232.0	91 **Pa** (231)	92 **U** 238.0	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (260)

List of Selected Constant Values

Speed of light in a vacuum	c	=	3.0×10^{8} m s^{-1}
Avogadro's number	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant	F	=	9.65×10^{4} C mol^{-1}
Planck constant	h	=	6.6256×10^{-34} J s
Reduced Planck constant	$\hbar$	=	1.054×10^{-34} J s
Rydberg constant	R_H	=	1.097×10^{7} m^{-1}
		=	2.18×10^{-14} J
Molar of gases constant	R	=	8.314 J K^{-1} mol^{-1}
		=	0.08206 L atm mol^{-1} K^{-1}
Boltzmann constant	k	=	1.3807×10^{-27} J K^{-1}
Mass of proton	M_p	=	1.672×10^{-27} kg
Electronic Bohr magneton	μ_B	=	9.2741×10^{-24} J T^{-1}
Nuclear Bohr magneton	β_N	=	5.05×10^{-27} J T^{-1}
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	1.602×10^{-19} C

Unit and Conversion Factor

Energy	1 J = 1 kg m^{2} s^{-2} = 1 N m = 10^{7} erg
	1 calorie = 4.184 Joule
	1 eV = 1.602×10^{-19} J
	1 amu = 1.66×10^{-27} kg
Pressure	1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101325 N m^{-2}